

University Of Education
Jauharabad Campus

Assignment 6

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Department: BSCS-(6th-Morning)
Subject Title: Compiler Construction

SLR(1) Parser

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Example # 01 :-

Grammar $\rightarrow$

$$E \rightarrow E + \bar{T} / \bar{T}$$

$$\bar{T} \rightarrow \bar{T} * F / F$$

$$F \rightarrow (E) / id$$

Step # 01 :-

Augment the Grammar & Number it.

$$E' \rightarrow E^{(0)}$$

$$E \rightarrow E + \bar{T}^{(1)} / \bar{T}^{(2)}$$

$$\bar{T} \rightarrow \bar{T} * F^{(3)} / F^{(4)}$$

$$F \rightarrow (E)^{(5)} / id^{(6)}$$

Step # 02 :-

First & Follow.

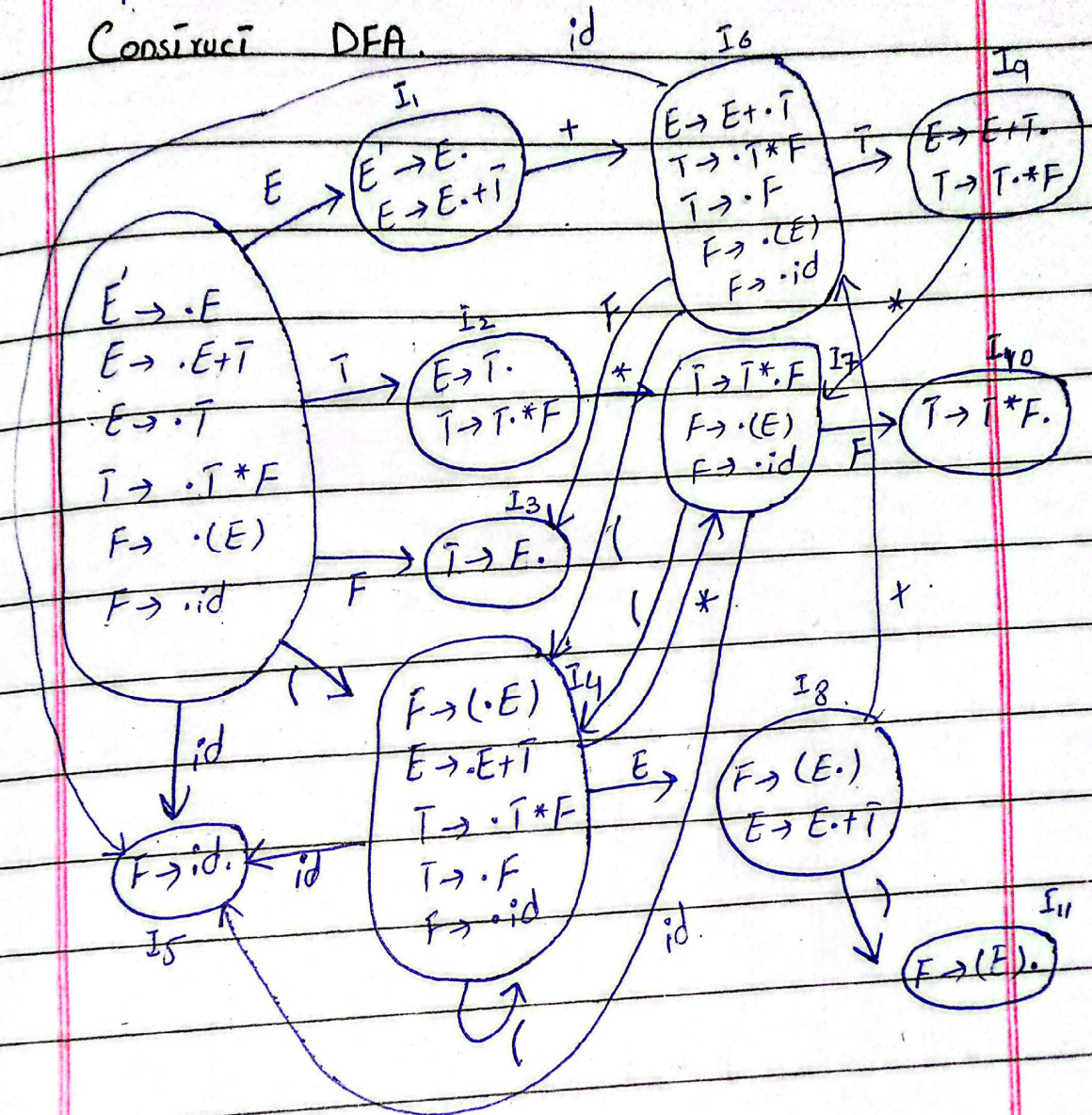
Grammar	First	Follow
$E' \rightarrow E$	$\{ (, id \}$	$\$$
$E \rightarrow E + \bar{T} / \bar{T}$	$\{ (, id \}$	$\$, +,)$
$\bar{T} \rightarrow \bar{T} * F / F$	$\{ (, id \}$	$\$, +,), *$
$F \rightarrow (E) / id$	$\{ (, id \}$	$\$, +,), *$

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Step # 03 :-

Construct DFA.



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Step # 04 - Parsing table.

State	Action						Goto		
	+	*	(	)	id	\$	E	T	F
I ₀			S ₄		S ₅		1	2	3
I ₁	S ₆					Accept			
I ₂	r ₂	S ₇		r ₂		r ₂			
I ₃	r ₄	r ₄		r ₄		r ₄			
I ₄			S ₄		S ₅		8	2	3
I ₅	r ₆	r ₆		r ₆		r ₆			
I ₆			S ₄		S ₅		9	3	
I ₇			S ₄		S ₅				10
I ₈	S ₆			S ₁					
I ₉	r ₁	S ₇		r ₁		r ₁			
I ₁₀	r ₃	r ₃	r ₃	r ₃	r ₃	r ₃			
I ₁₁	r ₃	r ₅	r ₅	r ₅		r ₅			

$$I_2 \# I_2 = F \rightarrow T$$

$$I_3 \# T \rightarrow F$$

$$\{ \$, +,), \}, 2 = r_2$$

$$[\$, +, *,), 3] = r_4$$

$$I_5 \# I_5 = F \rightarrow id$$

$$I_9 \# E \rightarrow E + T$$

$$\{ +, *,), \$ \} 5 = r_6$$

$$+,), \$, 9 = r_1$$

$$I_{10} \# T \rightarrow T * F$$

$$I_{11} \# F \rightarrow (E)$$

$$+, *,), \$, 10 = r_3$$

$$\{ +,), *, \$, 1 \} = r_5$$

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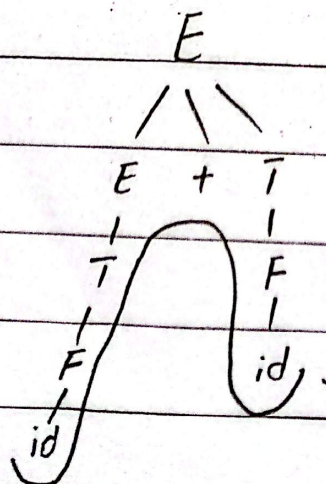
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Processing string #
id + id \$

Step # 05 - Stack Implementation.

Stack	Input	Action
0	id + id \$	Shift id $\rightarrow$ S
0 id \$	+ id \$	reduce $F \rightarrow id$
0 F \$	+ id \$	reduce $T \rightarrow F$
0 T \$	+ id \$	reduce $E \rightarrow T$
0 E \$	+ id \$	Shift + $\rightarrow$ S
0 E + \$	id \$	Shift id $\rightarrow$ S
0 E + id \$	\$	reduce $F \rightarrow id$
0 E + id F \$	\$	reduce $T \rightarrow F$
0 E + T \$	\$	reduce $E \rightarrow F + T$
0 E \$	\$	Accept

Step # 06 $\rightarrow$ Parse Tree.



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Example # 02 :-Grammar $\# \rightarrow$

$$S \rightarrow \lambda = R / R$$

$$L \rightarrow *R / id$$

$$R \rightarrow \lambda$$

Step # 01 $\# \rightarrow$ Augment Grammar Σ
Number it.

$$S' \rightarrow S \text{ ①}$$

$$S \rightarrow L \overset{\text{①}}{=} R \overset{\text{②}}{/} R$$

$$L \rightarrow \overset{\text{③}}{*} R \overset{\text{④}}{/} id$$

$$R \rightarrow L \text{ ⑤}$$

Step # 02 $\#$.

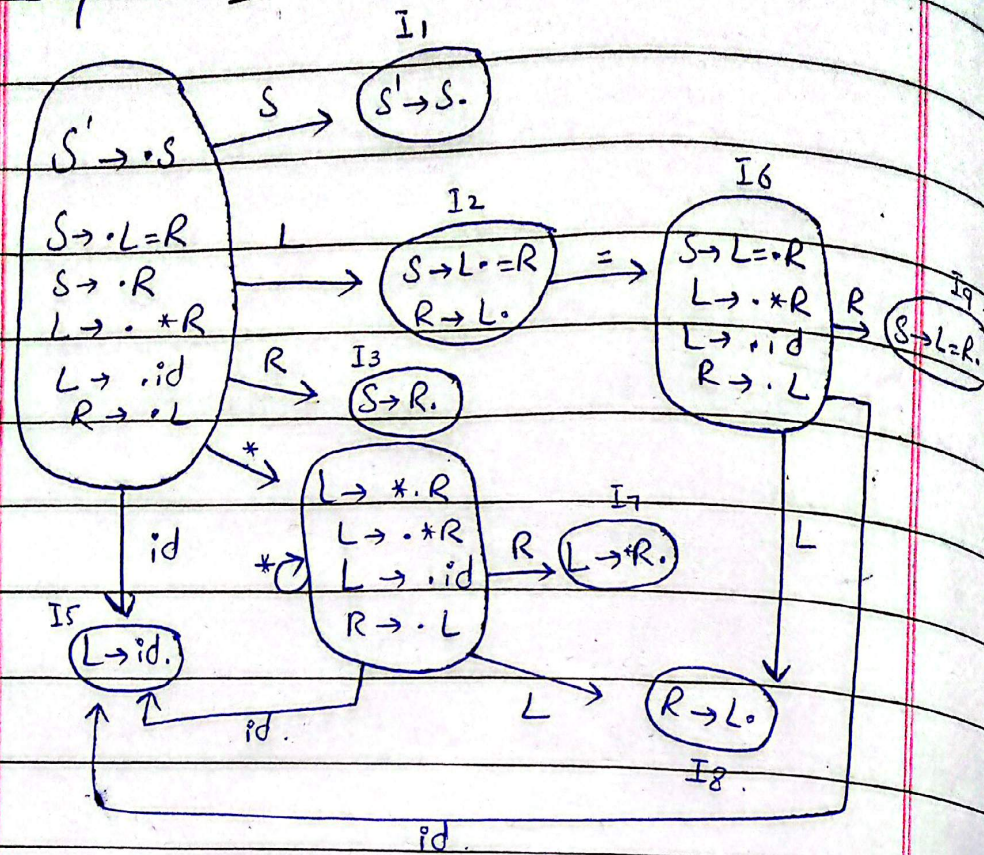
First ϵ Follow.

Grammar	First	Follow.
$S' \rightarrow S$	$\{ *, id \}$	$\{ \$ \}$
$S \rightarrow L = R / R$	$\{ *, id \}$	$\{ \$ \}$
$L \rightarrow *R / id$	$\{ *, id \}$	$\{ =, \$ \}$
$R \rightarrow L$	$\{ *, id \}$	$\{ *, \$ \}$

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Step # 03: Construct DFA.



Step # 04: Parsing Table. #

State	Action				Goto		
	=	*	id	\$	S	L	R
I_0		s_4	s_5	$\$$	1	2	3
I_1				Accept.			
I_2	s_6, r_5			r_5			
I_3				r_2			
I_4		s_4	s_5				7
I_5	r_4			r_4			
I_6		s_4	s_5			8	9
I_7	r_3			r_3			
I_8	r_5			r_5			
I_9				r_1			

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 $I_2 \# R \rightarrow L. \quad ; \quad F(R) = \{ =, \$ \} : 2$
 $\rightarrow \{ =, \$ \} = \text{reduce } 5.$
 $I_3 \# S \rightarrow R. \quad ; \quad \{ \$, 3 \} = \text{reduce } 2$
 $I_5 \# L \rightarrow id.$
 $\{ =, \$, 5 \} = \text{Reduce } 4$
 $I_7 \# L \rightarrow *R. \quad ; \quad \{ =, \$, 7 \} = \text{Reduce } 3$
 $I_8 \# R \rightarrow L. \quad ; \quad \{ =, \$, 8 \} = \text{Reduce } 5$
 $I_9 \# S \rightarrow L = R. \quad ; \quad \{ \$, 9 \} = \text{reduce } 1$
 $### \rightarrow$

* The Grammar contains a Shift reduce conflict in state I_2 on symbol '=', therefore it is not SLR(1)

* Hence, deterministic stack based LR Parsing implementation cannot be performed unless Grammar is modified or more powerful parser is used.